

DIRECTIONS for questions 1 to 4: The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Turtles may never take top place among the animal kingdom's most prolific vocalizers, but it turns out that they do indeed have something to say. In a new study published in *Nature Communications*, researchers found that turtles, along with other understudied animals, do in fact communicate using a diverse repertoire of vocal sounds. The study's authors suggest that their finding may push the origins of acoustic communication back in time to the common ancestor of all lunged vertebrates.

Prior to the current study, many of the included species "were considered to be mute," Gabriel Jorgewich-Cohen, a doctoral candidate at the University of Zurich, tells *Scientific American*. By listening carefully to recordings from 53 species the team reached a different conclusion: that vocalization is more widespread than previously thought, and that "the sounds that turtles are making have the same evolutionary origin as our own vocal communication," Jorgewich-Cohen says.

The paper's findings add fresh fuel to debates around the abilities of some animals to communicate with one another. In 2020, two scientists published a paper in *Nature Communications* in which they mapped the evolutionary phylogenies of roughly 1,800 vocal and non-vocal species and postulated that acoustic communication had evolved independently in Earth's major lineages (including frogs, birds, and mammals) in association with nocturnal lifestyles. In that analysis, turtles were lumped into the non-vocal group.

...Jorgewich-Cohen began to probe for sound in previously understudied species by studying his own pet turtles. "I decided to record them, just to check it out," he tells *New Scientist*. "I found several sounds there, and then we just kept going [with more species]. And suddenly, I had good sampling, and I could understand a bigger picture."

From there, the team collected sounds from an additional 50 turtle species, as well as lungfish, tuatara, and caecilians. To better identify which scenarios might elicit sound, Jorgewich-Cohen traveled to five countries to record each species for at least 24 hours, and did so in various settings, including when the animals were alone or in same- or mixed-sex groups, and even when they were underwater. Every species the group studied produced at least one sound, and in many cases, these recordings were the first time such sounds had ever been heard.

While the study adds to scientists' understanding of vocalizations in these groups, it also has implications for the evolution of auditory communications more broadly. When the researchers reanalyzed the earlier phylogenies with their data added in, they concluded that, rather than evolving multiple times, vocalization evolved once in a common ancestor. Specifically, Jorgewich-Cohen and colleagues traced vocalization back to the lobe-finned fish *Eoactinistia foreyi*, which is considered a possible last common ancestor of all choanate (lunged) vertebrates. That would mean that vocal communication evolved roughly 407 million years ago, at least 100 million years earlier than previously thought.

Jorgewich-Cohen tells *New Scientist* that vocalization may be even older still, as lungless fish also produce sounds. "It could be that one lineage of those fishes was the precursor of the type of sound that we make as [choanates]," he says. "So, it could be actually that this lineage of sound production is older than what I found."

Q1. The ideas in the passage are BEST represented by which of the following sequences?

- ☐ a) Vocal communication—acoustic communication—turtles—choanates—lungless fish
- ☐ b) Nocturnal lifestyles—turtle vocalisation—choanate auditory communication—lungless fish communication
- ☐ c) Turtle vocalisation—animal communication—auditory communication in lunged vertebrates—vocalisation in lungless fish
- ☐ d) Animal lineages—evolutionary origin of animal communication—phylogenies—lobe-finned fish—lungless fish

Q2. Which of the following best restates the 'implications' in "it also has implications for the evolution of auditory communications more broadly" (para 6)?

- ☐ a) Vocalisation was traced all the way back to the last common ancestor of all lunged vertebrates.
- ☐ b) Vocal communication evolved with lungless fish.
- ☐ c) All lunged vertebrates have a common ancestor.
- ☐ d) Vocalisation didn't evolve independently across various lineages.

Q3. The 2020 paper published in Nature Communications was mentioned to show that

- ☐ a) some animals were considered incapable of producing sounds earlier.
- ☐ b) turtles are incapable of vocal communication.
- ☐ c) acoustic communication had evolved independently in Earth's major lineages.
- ☐ d) nocturnal lifestyles influenced the vocalisation abilities of various lineages.

Q4. Jorgewich-Cohen travelled to five countries to record each species based on the presumption that the recorded species

- ☐ a) do not always produce the same sounds in the same setting.
- ☐ b) do not always produce sounds.
- ☐ c) produce different sounds in different settings.
- ☐ d) produce different sounds when alone and when in groups.

DIRECTIONS *for questions 5 to 8:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Guerrilla warfare, occurring between lightly armed partisans and a conventional army, is an example of asymmetrical warfare. Victory in war does not always go to the militarily superior force. Indeed, colonial powers have contended with asymmetrical threats since the rise of empires. In the 6th century BCE, Darius I of Persia, at the head of the largest and most powerful army in existence at the time, was checked by the Scythians, who possessed a smaller but far more mobile force. As recounted by Herodotus in Book IV of his History, the Scythians retreated before the main body of the Persian army, drawing it deeper into Scythian territory, only to launch lethal mounted strikes on Persian encampments. Darius was forced to retire, leaving the Scythians in command of the lands beyond the Danube River.

In the modern era, Western powers fighting in developing countries have sometimes been defeated by local forces despite massive asymmetries in terms of conventional military strength. Colonial powers were forced to withdraw from Algeria, Indochina, and other areas not necessarily as a result of defeat in battle but because of their lack of will to sustain the war. In Vietnam a crushing defeat at the Battle of Dien Bien Phu in 1954 sapped the will of the French military, and, after some two decades of U.S. involvement in the Vietnam War, the social and political environments at home forced the United States to concede defeat and withdraw its forces. The insurgents in colonized countries often did not need to defeat the sometimes long-established colonizer but merely persuaded it to withdraw from the region. Asymmetries of both power and will were operating: the colonial powers possessed superior military resources but were sometimes reluctant or unable to bring them to bear.

The value of asymmetrical tactics can be seen most clearly in guerrilla warfare – indeed, guerrilla means “little war” in Spanish. Guerrilla fighters are generally fewer in number and possess fewer and less-powerful weapons than the opposing force. Guerrilla tactics include ambush, avoiding open battle, cutting communication lines, and generally harassing the enemy. Guerrilla warfare has been practiced throughout history, and it includes both military operations carried out against the rear of an enemy’s army and operations carried out by a local population against an occupying force. The aim of the guerrilla fighter is erosion of the enemy’s will to sustain the costs of continuing the war. Henry Kissinger observed that “the guerilla wins if he does not lose. The conventional army loses if it does not win.”

Although usually exercising a smaller force, guerrilla fighters, especially in urban areas, can be formidable adversaries to a conventional military. Guerrilla fighters typically do not inhabit large, well-established bases, making it impossible for their enemy to exploit technological advantages such as aerial bombardment to destroy personnel and infrastructure. If the guerrillas are in an urban area, their opponents cannot use powerful conventional weapons unless they are willing to inflict large numbers of civilian casualties and risk increasing popular support for the guerrillas. Small guerrilla or insurgent groups also tend to be less hierarchical, meaning that a force cannot be neutralized by the capture or death of a handful of leaders.

Q5. All of the following are true according to the passage EXCEPT:

- ☐ a) Algeria, Indochina and Vietnam serve as examples for the effectiveness of asymmetrical warfare.
- ☐ b) In the 6th century BCE, the Scythians had to concede defeat due to the Persian army’s tactics.
- ☐ c) Guerilla warfare includes covert operations against enemy forces.
- ☐ d) Guerillas may benefit from the disinclination of conventional armies to attack urban areas.

Q6. The purpose of the second para of the passage is to show that

- ☐ a) there are cases in which conventional military strength has preferred armistice in the face of long-drawn asymmetrical wars.
- ☐ b) the US lost the Vietnam war because of the asymmetrical war launched by the local forces.
- ☐ c) the US and French fought the Vietnam War together.
- ☐ d) the social and political climate of Vietnam pressured the US to concede defeat in the Vietnam war.

Q7. All of the following are implied by the penultimate para EXCEPT that conventional armies

- ☐ a) are under pressure to win the wars they are fighting.
- ☐ b) are constantly wary of the mounting costs of continuing a war.
- ☐ c) do not prefer wars of attrition that do not give results in reasonable time frames.
- ☐ d) do not prefer engaging guerilla fighters because it always includes local populations.

Q8. The example of the Scythians helps us understand that

- ☐ a) smaller forces are more efficient than larger armies.
- ☐ b) guerilla warfare was employed since the rise of empires.
- ☐ c) asymmetrical warfare has been effective against colonial powers since times immemorial.
- ☐ d) smaller mobile forces have troubled larger armies as far back as in the 6th century BC.

DIRECTIONS *for questions 9 to 12:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

Diamond, with its tough-to-break carbon lattice of interlocking cubes, is traditionally considered the hardest material on Earth. Yet a rare form of diamond known as lonsdaleite – a crystal with carbon atoms arranged in flexing three-dimensional hexagons – may be even harder. To date, natural lonsdaleite has been found only in impact craters, where it has formed by the intense pressure of meteorites crashing to Earth. But now researchers say that they've identified lonsdaleite crystals that formed billions of years before the meteorites carrying them ever reached the planet.

... The research team examined 18 different meteorite samples from a family known as "ureilites." Because ureilites are relatively homogeneous in their chemical composition – which is uniquely rich in carbon – scientists theorize they originate from the same parent body. "There was this dwarf planet just after the start of our solar system – so 4.5 billion years ago – and the planet got hit by an asteroid," says Alan Salek, a graduate student in applied physics at the Royal Melbourne Institute of Technology in Australia and co-author of the new study. This cataclysmic impact tore the dwarf planet apart, sparking a chemical reaction that could have turned pieces of the planet's graphite into lonsdaleite, he adds.

Graphite is made up of flat layers of carbon atoms bonded together as hexagons. These stacked layers are weakly attracted to each other and relatively easy to pull apart. On Earth, high heat and pressure can rearrange these carbon atoms into a 3-D lattice of cubes, thereby creating the traditional kind of diamond. But a brief period of extremely intense pressure – such as that of a meteorite impact – can preserve graphite's original hexagonal arrangement while its layers bond into the strong 3-D lattice of lonsdaleite.

The researchers propose that rather than the rapid impact pressure known to produce tiny lonsdaleite crystals on Earth, these samples instead formed through a rapid release of pressure. They claim that a fluid mix of carbon, hydrogen, oxygen and sulfur was heated and pressurized in the dwarf planet's mantle until an asteroid impact smashed that mantle into pieces. Study co-author Andrew Tomkins says that the rapidly depressurizing mix of chemicals could have interacted with the dwarf planet's graphite to transform it into lonsdaleite.

In this particular reaction, graphite crystals would have been essentially torn apart and rebuilt into lonsdaleite. "It's called 'coupled dissolution-precipitation' because it's kind of dissolving this thing and replacing it at the same time," Tomkins says. This fluid-driven reaction took place in chunks of the dwarf planet as they went flying into space... Tomkins explains that the structure of these meteorites' minerals indicates a rapid cooling process that points to a dramatic collision. Looking at particular radioactive signatures of the minerals, the researchers estimated a date for this collision – roughly 4.5 billion years ago. Plus, the ureilite samples contain interlocking layers of lonsdaleite, cubic diamond and graphite in a pattern that points to the fluid-driven transformation Tomkins's team describes. The specks of lonsdaleite that Salek and Tomkins' research group identified were up to a micron in size – still extremely small but roughly 1,000 times larger than any lonsdaleite crystals previously known. This suggests that a fluid-driven transformation of graphite into lonsdaleite might produce bigger crystals than the impact method...

Q9. Which of the following is not true about the difference between lonsdaleite crystals previously known and the ones researched by Salek and Tomkins' research group?

- ☐ a) The former are much smaller than the latter.
- ☐ b) The former are produced by rapid impact pressure whereas the latter were produced by rapid release of pressure.
- ☐ c) The former have a different crystal structure compared to the latter.
- ☐ d) The former were produced on earth whereas the latter were produced on a different planet.

Q10. 'Ureilites' provided a distinct advantage to the research owing to the fact that:

- ☐ a) All of them had a similar chemical composition.
- ☐ b) All of them were formed by the intense pressure of meteorites crashing into earth.
- ☐ c) All of them have the same amount of carbon.
- ☐ d) All of them have carbon atoms arranged in flexing three dimensional hexagons.

Q11. Which of the following, if true, most strengthens the research mentioned in the passage?

- ☐ a) The dwarf planet's mantle had all the essential chemicals needed for the fluid driven reaction that created lonsdaleite.
- ☐ b) The asteroid that hit the dwarf planet had graphite present in it.
- ☐ c) The asteroid that hit the dwarf planet generated sufficient heat to create the fluid mixture of carbon, oxygen, hydrogen and sulphur.
- ☐ d) Chunks of the dwarf planet became part of the earth's core after the asteroid impact.

Q12. Which of the following is NOT true about the 'coupled dissolution-reprecipitation' that created lonsdaleite as proposed by the research discussed in the passage?

- ☐ a) It helped convert graphite crystals into lonsdaleite.
- ☐ b) It was triggered by the probable collision of an asteroid with a dwarf planet.
- ☐ c) It was catalysed by the rapid impact pressure needed to create lonsdaleite crystals.
- ☐ d) It involved a fluid mix of carbon, hydrogen, oxygen and sulphur.

DIRECTIONS *for questions 13 to 16:* The passage given below is accompanied by a set of **four** questions. Choose the best answer to each question.

So how many classical traditions might there be? By my reckoning, at least 15, and though they seem diverse, they coalesce into a handful of stylistic families. Gong-chime music links Thailand, Laos, and Cambodia with Indonesia, while the music of China's Tang dynasty was imported by Japan as its court music, a function which it still performs today. Meanwhile it's a vertiginous thought that European classical music and the music of the Muslim Middle East should have a common root in classical Greek theory, which itself derived from ancient Babylon. Translated by Arab theorists in the eighth and ninth centuries, the writings of the Pythagoreans became the underpinning for the Arabic "science of music" which laid down rules for the structure of scales and the mathematical tuning of intervals; versions of this were then absorbed by the medieval church.

Yet a music can die, as a language can. Its death may be a natural consequence of evolutionary change, but it may also reflect the abrupt fall of an empire, or result from persecution: the Taliban lost no time in driving Afghanistan's entire classical-music community into exile; Malian musicians are currently in the firing line for local Islamists.

A bigger threat comes via insidious erosion by the homogenising tide of western music, both commercial and classical. At the start of the last century, musicians across a swath of the globe stretching from Egypt to Japan were led by their rulers to believe that their indigenous monophonic styles were "backward" (only India was immune from this pernicious notion), and that, in order to compete on the international stage, they needed to deploy western polyphony. Microtonal scales, with their infinite variety, were ironed out in favour of European diatonic scales, and intimate solo instruments such as the Kazakh dombra lute found themselves grotesquely multiplied in the Soviets' "folk" orchestras.

Today, armies of conservationists are rushing to the rescue. UNESCO has designated a long list of musics as "intangible cultural heritage"; the Aga Khan Trust for Culture has set up a network of schools throughout central Asia where master-musicians now pass on their skills to young performers. Cylinder and shellac recordings are being studied by Middle Eastern musicians seeking to revive forgotten styles; governments are realising the value of traditional music as a propaganda tool.

But no music can survive long on artificial life-support: without a driving social impulse, it's just a piece of museum culture, and this could well be the fate of the Uzbek-Tajik shashmaqom – a fusion of vocal and instrumental music, melodies, rhythms and poetry – whose continued existence now depends, despite its venerable history, on the enthusiasm of a handful of musicologists.

All of which raises a big question: if some classical musics go to the wall, will they be replaced by new ones? Not necessarily. Indeed, given the social conditions required for their gestation, and given the exponential rate at which the geopolitical order is now mutating, it's hard to imagine new classical forms emerging anywhere in the foreseeable future. Folk musics will continue to burst forth as they always have done. But with classical music, what we have may be all we'll get, so we should treasure it.

Q13. Which of the following, if true, contradicts any of the author's arguments in the passage?

- ☐ a) The proliferation of classical forms of music is hampered by homogenising musical elements around the world.
- ☐ b) Some musical forms have the capacity to drive social impulse and thrive on it.
- ☐ c) Not all musical forms are on the verge of extinction considering they are practiced by large populations.
- ☐ d) Music and language differ in the sense that the former entertains whereas the latter is a practical necessity.

Q14. It can be inferred from the third para of the passage (“A bigger threat...Soviets’ “folk” orchestras.”) that:

- ☐ a) Indigenous classical forms of music in India had encouragement from rulers.
- ☐ b) Indian monophonic styles resisted the homogenising tide brought by global musicians.
- ☐ c) Indian music boasts of the infinite variety present microtonal scales.
- ☐ d) Solo instruments produce scales that grotesquely adulterate monophonic styles.

Q15. Which of the following best restates why the author refers to 'artificial life-support' in the penultimate para of the passage?

- ☐ a) Many steps are being taken by both the government and the musicians to conserve and protect classical forms of music.
- ☐ b) The enthusiasm of only a handful of musicologists cannot really conserve classical forms of music.
- ☐ c) Classical forms of music can thrive only with social engagement and not just with academic and government interest.
- ☐ d) Classical music cannot be conserved without the active participation of music conservationists.

Q16. Which of the following most helps us evaluate the author's answer to 'a big question' mentioned in the last paragraph of the passage?

- ☐ a) Some classical forms have a smaller gestation period than others.
- ☐ b) The mutation of geopolitical orders results in unsettling of cultural norms of various populations.
- ☐ c) Folk music is gaining more popularity than classical music because of its catchiness.
- ☐ d) No classical form of music has originated in any part of the world in the last two centuries.

Q17. DIRECTIONS *for question 17:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

Variations in work environments greatly affect job demands even within individual industries and for fairly narrowly defined tasks. A good illustration of the practical difficulties this causes is provided by the experience of training officials in the British construction industry seeking to identify a set of common job-related competencies that could be used as a basis for defining industry-wide training needs. Job related know-how was found to be very dependent upon the context in which people worked: physical working conditions, size of construction site, size of firm and technology used. As a result, there was great disagreement among employers as to the know-how considered relevant even for fairly narrowly defined operations.

- ☐ a) Hiring for specific tasks could be difficult because of the disagreement between employers around what is the right criteria.
- ☐ b) Variations in work environments could make it hard to identify the competencies required to hire individuals for narrowly defined tasks.
- ☐ c) Job-related knowledge varies from industry to industry because of its dependence on work environments.
- ☐ d) Even inside the same industry, a variation in work environment influences the skills needed to carry out specific tasks.

Q18. DIRECTIONS *for question 18:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

- 1. As Pleasure Activism’s author writes, we need to make “justice and liberation the most pleasurable experiences we can have”.
- 2. Embracing collective joy may well rest, though, on vanquishing the punishing, singular idea of happiness; this singular idea is that it’s connected to efficiency, a person’s job or even our individual wellbeing.
- 3. Collective campaigns, however limited in their gains, can take ordinary, lonely geographies and transform them into spaces of hope.
- 4. Collective joy, in contrast, is an attempt to alchemize, say, loss or powerlessness into significance, through physical movement, sound and embodied politics.

Q19. DIRECTIONS *for question 19:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

It turns out that the surface of the moon is full of mysteries, particularly its fine dust covering. In the lunar soil, scientists hope to find clues about the formation of our solar system and glean answers about what other worlds might be like. It is also the destination for a new generation of crewed missions and the most likely site for the first permanent off-world human outpost. So not only will we want to learn how to live with this vexing substrate, we're also going to need to learn how to build with it. In other words, we're going to have to get our hands dirty.

- ☐ a) The fine dust on the lunar surface has to be studied to understand our solar system, other worlds, and how to build a permanent human outpost.
- ☐ b) The surface of the moon has the answers to how the solar system and other worlds were formed.
- ☐ c) We have to understand the surface of the moon if we want to live on it and build on it using its own material.
- ☐ d) Crewed missions are studying the lunar surface to understand how the solar system and other worlds were formed so future outposts can be built on it.

Q20. DIRECTIONS *for question 20:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. Unlike the opulent thobes elsewhere in the gallery, it's a thoroughly worn garment created for everyday wear.

2. As with all of the embroidered clothes and items currently on view, it's an expression of class, gender and sociopolitical change.

3. But it's also a fascinating window into the life of a rural woman from the Gaza region in the 1930s: the knee patches replace fabric worn away from field and domestic work, while the gaps by the chest panel show alterations made for breastfeeding.

4. With its faded fabric and darned-up holes, the first Palestinian dress on display at Kettle's Yard in Cambridge might initially seem unremarkable.

Q21. DIRECTIONS *for question 21:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: If you say the phrase “economic theory,” people know what you mean.

Paragraph: _____(1)_____ Economics is considered the most powerful of the social sciences in an intellectual sense. That power derives from the fact that economics has a unified, core theory from which nearly everything else follows. _____(2)_____ No other social science has a similar foundation. _____(3)_____. Rather, theories in other social sciences tend to be for special purposes – to explain what happens in a particular set of circumstances. In fact, economists often compare their field to physics; like physics, economics builds from a few core premises. _____(4)_____

☐ a) Option 1

☐ b) Option 2

☐ c) Option 3

☐ d) Option 4

Q22. DIRECTIONS *for question 22:* The paragraph given below is followed by four summaries. Choose the option that best captures the essence of the paragraph.

Anticolonial movements should not be viewed as isolated events; indeed, many anticolonial leaders and organizations learned from other movements. Ania Loomba, an English professor, noted that there were important political and intellectual exchanges between different anticolonial movements and individuals and that even the most rooted and traditional of these was shaped by a syncretic history. Many of the classic writings associated with anticolonial movements continue to hold salience in contemporary society. This continuity is testimony to the powerful ideas that embraced anticolonial movements, namely concerns with sovereignty, equality, and social justice.

- ☐ a) Anticolonial movements have a shared history that still influences contemporary society.
- ☐ b) Anticolonial movements were ideologically connected to each other, and those ideologies are still relevant in contemporary times.
- ☐ c) Anticolonial movements inspired each other and also inspired writings that are still relevant for their ideas in contemporary society.
- ☐ d) Anticolonial movements inspired classic writings about powerful ideas like sovereignty, equality and social justice.

Q23. DIRECTIONS *for question 23:* The sentences given in the question, when properly sequenced, form a coherent paragraph. Each sentence is labelled with a number. Decide on the proper order for the four sentences and key in the sequence of four numbers as your answer, in the input box given below the question.

1. But before the publication of Myrdal's *An American Dilemma*, with its concise review of studies of "racial differences" between whites and blacks, many a sociologist would leave the door open for biologically based race theories, while the Frankfurt School excluded race theories from explaining anything.
2. As social scientists, we are so familiar with the negative heuristics of the authoritarian character research program that it is at first difficult to see it at all.
3. This was not the case during the 1920s and 1930s, notwithstanding the fact that the Nazi era escalated the use of biological, social Darwinist theories in an unprecedented way and produced absurd racist theories.
4. That neither genetics nor "race" can explain the character structure of persons, or their psychic problems and disorders is the common sense of the social sciences and of neurobiology today.

Q24. DIRECTIONS *for question 24:* There is a sentence that is missing in the paragraph below. Look at the paragraph and decide in which blank (option 1, 2, 3, or 4) the following sentence would best fit.

Sentence: Immunology and neurosciences have since grown into rich and complex disciplines.

Paragraph: Neurosciences have a long history, tracing back to antiquity, in the quest for understanding the senses, the mind, and the brain. ____ (1) _____. Immunology, in contrast, was born at the end of the 19th century from the necessity to understand how we survive microbes, just shown to be the source of infectious diseases. ____ (2) _____. The 19th century also gave rise to neuroimmunology, largely to unravel the mechanisms driving pathologies of the nervous system, an aim still shared by most clinical neuroscience departments. ____ (3) _____. Nevertheless, only on rare occasions have the fields crossed paths to reveal how the two systems crosstalk to maintain health, regulate physiology and alter behavior. ____ (4) _____

☐ a) Option 1

☐ b) Option 2

☐ c) Option 3

☐ d) Option 4